Rajdeep Gill

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Education

University of Manitoba — Winnipeg, MB
Bachelor of Science in Computer Engineering — *Sept. 2021 - May 2026*

- GPA: *4.45/4.5*
- Relavent Coursework: *Engineering Algorithms, Data structures and Algorithms, Systems Engineering Principles*

Experience

Undergraduate Research Student — May 2024 - August 2024
University of Manitoba - Biosensors Research Lab — *Winnipeg, MB*

- Designed and trained a neural network to predict biological cell viability from cell diffraction patterns, achieving an accuracy of 85%, demonstrating the feasibility of non-invasive cell viability prediction.
- Enhanced particle diffraction centroid detection using computer vision, boosting success rate from 75% to 92%, allowing for better statistical analysis.
- Developed an interactive interface for data analysis, allowing for real time visualization of diffraction patterns.

Undergraduate Research Student — May 2023 - August 2023
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- Employed dimensionality reduction techniques, including t-SNE and principal component analysis to compare and cluster datasets, identifying potential correlations and directing future research directions.
- Utilized parallel processing to reduce run times, enhancing research team productivity.
- Streamlined data collection for a seperate lab project by developing a program to interface with an oscilloscope.

Projects

Discord Clone | *React, TypeScript, HTML/CSS, Next.js*

- Developed a feature-rich Discord Clone with real-time messaging using web sockets.
- Utilized Tailwind CSS to create a visually appealign and responsive user interface.
- Implemented API services to handle user authentication, voice and video calls.
- Provided users with extensive control, allowing them to edit and delete messages, create and delete servers.

Microprocessor Communication System | *C, Java*

- Designed and implemented a server using the TCP/IP stack in C, handling multiple concurrent connections.
- Developed a Java-based proxy server to interface between an Android application and the microprocessor server.
- Designed the Android application to allow users to control the LEDs, monitor the temprature and button states of the microprocessor.

Chess Game | *Python*

- Created a command line chess game using Python.
- Implemented features such as move validation, checkmate detection, and piece promotion.
- Currently developing multiplayer functionality to enable real-time play between users.

Awards

NSERC Undergraduate Student Research Award — 2023, 2024
Presidents Scholarship — 2021-2024

Technical Skills

Languages: Python, Java, C, JavaScript, HTML/CSS, R
Frameworks: React
Developer Tools: Git, VS Code, Visual Studio, Android Studio, IntelliJ